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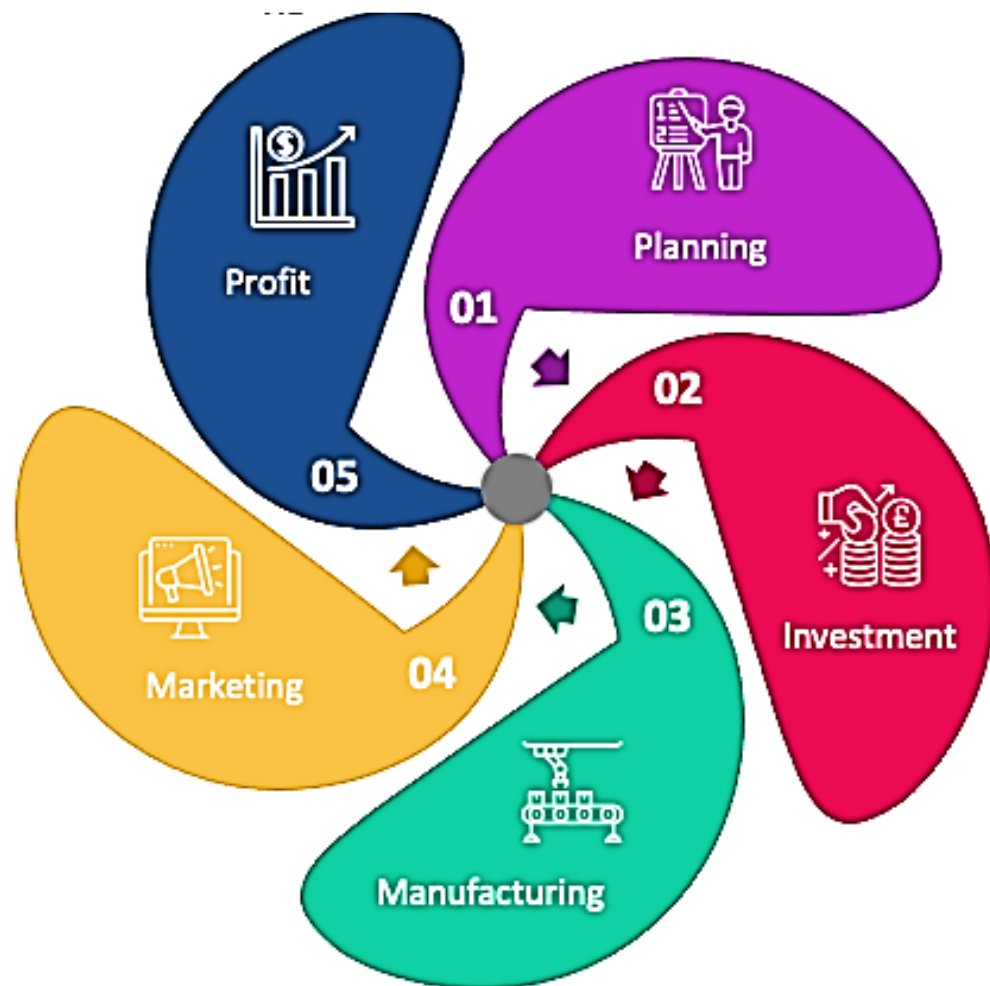
In the name of Allah, Most Gracious,  
Most Merciful.



# Course HS - 402, Engineering Economics, CR 1, (2+0)/week



## Engineering Economics (HS - 407)



### Course Instructor

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## **Course Outline & Reference Books**

**Join Google Classroom:**

**<https://classroom.google.com/c/NzYwNDU5NjkwNDg2?cjc=rk4peiq>**

**rk4peiq**



## **Course Outline & Reference Books**

### **Reference Books**

- 1. Engineering Economy by William G. Sullivan, Elin W. Wicks, C. Patrick Koelling**
- 2. Economics by Campbell R Cornell, Stanley L. Bruce**
- 3. Project Management, the managerial process by Erik W. Larson, Clifford F. Gray**
- 4. Karl E. Case's Principles of Economics (10th Edition)**

#### **Office Visiting Hours**

- ❖ **Office 08 MED,**
- ❖ **Phone: Ext 327/329**
- ❖ **Office Hours: 0830 - 1530 hrs.**
- ❖ **Lunch Break 1230 - 1330 hrs.**
- ❖ **Student Hours: 0930 - 1630 hrs. Tuesday**

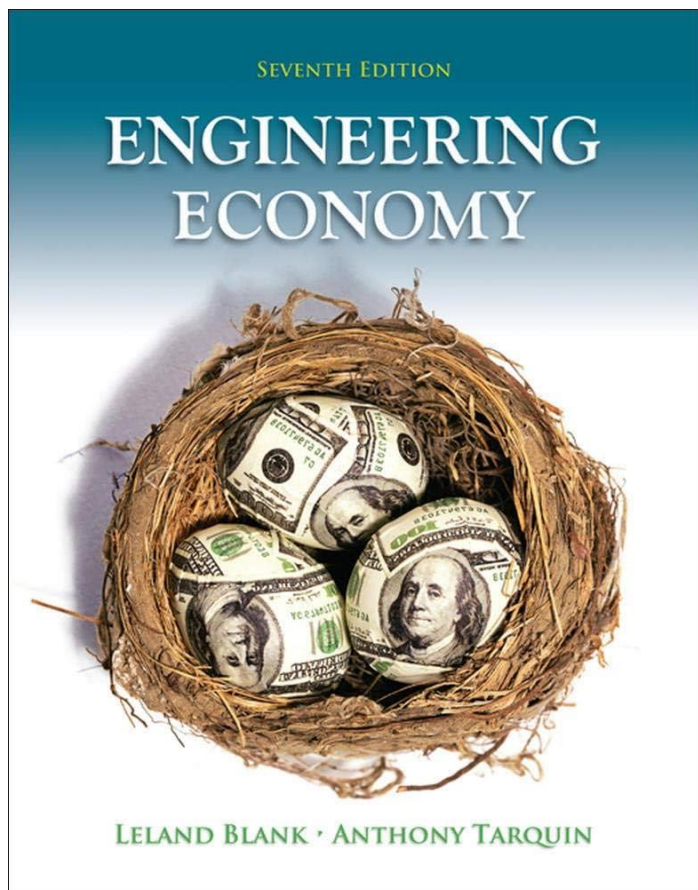


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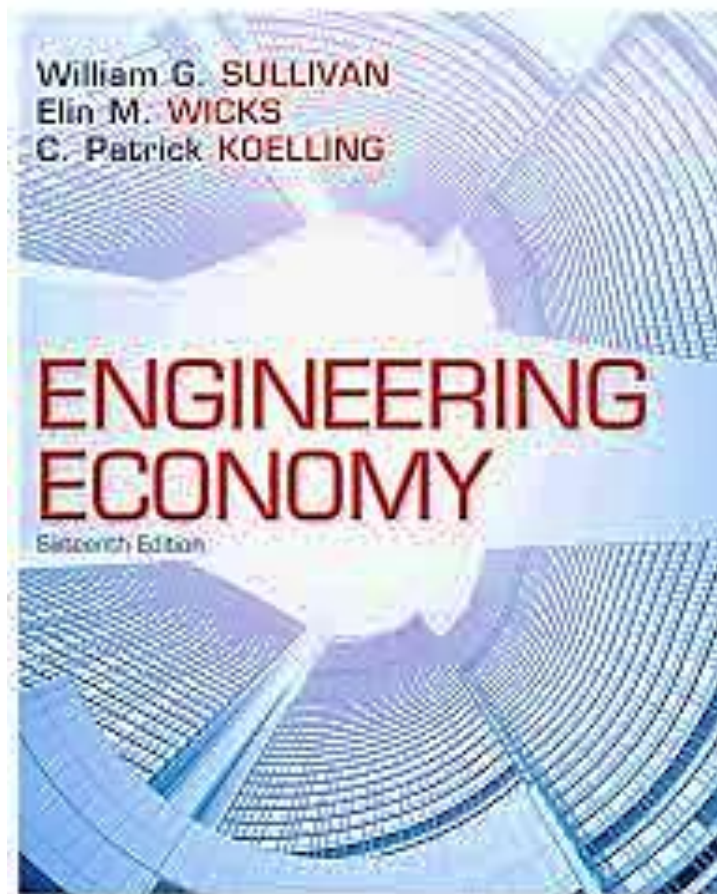


## Course Outline & Reference Books

### Reference Book 1



### Reference Book 2



### Reference Book 3





## Introduction to Engineering Economics (Time Value of Money)

**The time value of money (TVM) is a fundamental financial concept that says money available today is worth more than the same amount in the future, because of its potential to earn returns (like interest or investment gains).**

**If you have \$100 today, you can invest it and earn interest. That same \$100 a year from now isn't as valuable, because you've lost the opportunity to grow it.**

**Future Value (FV)**

$$FV = PV \times (1 + r)^t$$

**Present Value (PV)**

$$PV = \frac{FV}{(1 + r)^t}$$

**Future Value of an Annuity**

$$FV = PMT \times \frac{(1 + r)^t - 1}{r}$$

**Present Value of an Annuity**

$$PV = PMT \times \frac{1 - (1 + r)^{-t}}{r}$$

**Present Value (PV):** Value of money today.

**Future Value (FV):** Value of money at a future date.

**Interest Rate (r):** Growth rate per period.

**Time (t):** Number of periods (years, months, etc.).

**Payments (PMT):** Recurring payments, like annuities.



## **The Economic Problem: Compound Interest**

Aisha, a 28-year-old software engineer, inherited \$20,000 from her grandmother. Instead of spending it, she decided to invest the money in a special savings plan offered by her bank. The plan works in three phases:

**Phase 1 (1-5 Years):** Interest rate: 4% compounded annually. No withdrawals or additional deposits allowed.

**Phase 2 (6-10 Years):** Interest rate increases to 5% compounded quarterly. Aisha decides to deposit an additional \$1,000 every quarter during this period.

**Phase 3 (11-15 Years):** Interest rate changes to 6% compounded monthly. Aisha withdraws \$500 monthly to support her freelance business.

**How much money will Aisha have at the end of 15 years?**



## The Economic Problem: Compound Interest

### ✓ **Phase 1 (Years 1–5)**

Interest: 4% compounded annually

Principal: \$20,000

Time: 5 years

Formula:

$$A_1 = P \times (1 + r)^t = 20000 \times (1 + 0.04)^5$$

$$A_1 = 20000 \times (1.2166529) = \boxed{24,333.06}$$



## The Economic Problem: Compound Interest

### ✓ Phase 2 (Years 6–10)

New interest rate: 5% compounded quarterly

Time: 5 years = 20 quarters

Additional deposit: \$1,000 every quarter

Principal from Phase 1: \$24,333.06

Total after Phase 2:

$$A_2 = 31211.50 + 22563.00 = \boxed{53,774.50}$$

Step 1: Grow principal for 20 quarters

$$A_{\text{principal}} = 24333.06 \times (1 + 0.05/4)^{20} = 24333.06 \times (1.282037)$$

$$A_{\text{principal}} = \boxed{31,211.50}$$

Step 2: Add annuity for quarterly deposits

Annuity formula:

$$A_{\text{deposits}} = 1000 \times \left[ \frac{(1 + 0.05/4)^{20} - 1}{0.05/4} \right]$$

$$A_{\text{deposits}} = 1000 \times \left[ \frac{(1.282037 - 1)}{0.0125} \right] = 1000 \times 22.563 = \boxed{22,563.00}$$



## ✓ Phase 3 (Years 11–15)

## Compound Interest

New interest rate: 6% compounded monthly

Time: 5 years = 60 months

Monthly withdrawal: \$500

Principal from Phase 2: \$53,774.50

Final amount after 15 years:

$$A_3 = 72,538.67 - 34,885.00 = \boxed{\$37,653.67}$$

Step 1: Grow the principal

$$A_{\text{principal}} = 53774.50 \times (1 + 0.06/12)^{60} = 53774.50 \times (1.348850)$$

$$A_{\text{principal}} = \boxed{72,538.67}$$

Step 2: Subtract effect of monthly withdrawals

Withdrawal annuity formula:

$$A_{\text{withdrawals}} = 500 \times \left[ \frac{(1.06/12)((1 + 0.06/12)^{60} - 1)}{(0.06/12)} \right]$$

$$A_{\text{withdrawals}} = 500 \times \left[ \frac{(1.348850 - 1)}{0.005} \right] = 500 \times 69.77 = \boxed{34,885.00}$$



## **The Economic Problem: Compound Interest**

Zara, an aspiring entrepreneur, launched a coffee shop named Zara's Coffee Corner in January. To finance the business, she invested \$50,000 of her own savings and secured a \$30,000 bank loan. In the same month, she used \$25,000 to purchase coffee machines and furniture, paid \$6,000 for three months of rent in advance, and bought \$8,000 worth of initial inventory. In February, Zara earned \$12,000 in sales revenue but also incurred \$600 in utility expenses, paid \$3,000 in wages, and spent \$4,000 to restock inventory. In March, sales increased to \$18,000, while she again paid \$600 for utilities and \$3,000 for wages. Additionally, she paid \$500 in loan interest and withdrew \$2,000 for personal use. This case aims to evaluate Zara's financial position by applying the accounting equation— $\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$ —and by constructing a cash flow diagram over the three-month period to assess the inflows and outflows of cash.



## The Economic Problem: Compound Interest

### Outflows:

- Equipment:  $-\$25,000$
- Rent (prepaid for Jan–Mar):  $-\$6,000$
- Inventory:  $-\$8,000$  (Jan) +  $-\$4,000$  (Feb) =  $-\$12,000$
- Utilities:  $-\$600$  (Feb) +  $-\$600$  (Mar) =  $-\$1,200$
- Wages:  $-\$3,000$  (Feb) +  $-\$3,000$  (Mar) =  $-\$6,000$
- Interest Payment:  $-\$500$
- Owner Draw:  $-\$2,000$
- **Total Cash Outflows =  $\$52,700$**

### Inflows:

- Owner Investment:  $+\$50,000$
- Bank Loan:  $+\$30,000$
- Sales (Feb + Mar):  $+\$30,000$
- **Total Cash Inflows =  $\$110,000$**

### Cash Balance

$$\text{Net Cash} = 110,000 - 52,700 = \boxed{57,300}$$



## **The Economic Problem: Compound Interest**

| Asset                  | Amount          |
|------------------------|-----------------|
| Cash                   | \$57,300        |
| Equipment              | \$25,000        |
| Inventory (20% unsold) | \$2,400         |
| <b>Total Assets</b>    | <b>\$84,700</b> |

| Liability                    | Amount          |
|------------------------------|-----------------|
| Bank Loan (principal unpaid) | \$30,000        |
| <b>Total Liabilities</b>     | <b>\$30,000</b> |



## Owner's Equity



$$\text{Revenue} = 30,000 \quad ; \quad \text{Expenses} = 7,700 \quad ; \quad \text{Draw} = 2,000$$

$$\text{Net Income} = 30,000 - 7,700 = 22,300$$

$$\text{Equity} = 50,000(\text{initial}) + 22,300(\text{income}) - 2,000(\text{draw}) = \boxed{70,300}$$

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

$$84,700 = 30,000 + 54,700 \quad \leftarrow \text{Not balanced}$$

✂ Let's fix it: Since actual **Assets** = 84,700 and **Liabilities** = 30,000, the correct **Equity** must be:

$$\text{Equity} = 84,700 - 30,000 = \boxed{54,700}$$

So:

$$\text{Net Income} = 54,700 - 50,000 + 2,000 = \boxed{6,700}$$

$$\text{Hence, Total Expenses} = \text{Revenue} - \text{Net Income} = 30,000 - 6,700 = 23,300$$

This means earlier expense estimate (7,700) was missing some depreciation, rent consumption, or inventory adjustment.



**The Economic Problem: Compound Interest**

| Month | Cash Inflows | Cash Outflows | Net Cash Flow |
|-------|--------------|---------------|---------------|
| Jan   | \$80,000     | \$39,000      | +\$41,000     |
| Feb   | \$12,000     | \$7,600       | +\$4,400      |
| Mar   | \$18,000     | \$6,100       | +\$11,900     |
| Total | \$110,000    | \$52,700      | +\$57,300     |